MUHAMMAD ASIM

Senior Software Engineer · AI-Native Full-Stack Development
Lahore, Pakistan | +92 317 4332021 | asimm364@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Senior software engineer with **6+ years** shipping production systems across web, mobile, and cloud — now focused on **AI-native application development**. I build products *on top of* frontier models: integrating LLM APIs, wiring **RAG and vector search**, designing **agentic and tool-calling workflows**, and engineering prompts and context for reliable, structured output. I work daily in **Claude Code**, **Cursor**, and **Copilot**, using agentic coding tools to accelerate delivery while keeping review discipline and architectural judgment human-owned. Comfortable owning the full path from microservice architecture to CI/CD, observability, and latency and cost tuning of model-backed services.

CORE COMPETENCIES

AI Application Development: LLM API integration, Retrieval-Augmented Generation (RAG), vector search & embeddings, agentic workflows, function/tool calling, MCP (Model Context Protocol), prompt & context engineering, structured outputs, streaming, voice AI & conversational agents, AI video generation, guardrails & fallbacks, token/cost optimization

AI Coding Tools: Claude Code, Cursor, GitHub Copilot, Windsurf, Codex, v0, MCP servers — agentic pair-programming, spec-driven development, AI-assisted refactoring and code review

Model Providers & SDKs: Anthropic Claude, OpenAI (GPT-4/4o), Google Gemini, DeepSeek, Llama, Vercel AI SDK, LangChain, LangGraph, Pinecone / pgvector

Languages & Frameworks: TypeScript, JavaScript, Python, Ruby, Java, C++, SQL · React.js, Next.js, Node.js, Express, React Native, NestJS, GraphQL, Redux, Angular, Vue, Spring Boot, Ruby on Rails

Data & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Firebase · AWS (ECS, ECR, Fargate, EC2, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI

Distributed Systems: Microservices, Apache Kafka, Kafka Streams & Connectors, RabbitMQ, Pub/Sub, event-driven architecture

Web3 & Practices: Ethereum, ERC-20 & NFT flows, MetaMask, on-chain event verification · TDD, code review, observability, root-cause analysis, technical mentorship, agile delivery

EXPERIENCE

Senior Software Engineer · *SMS Holdings* Feb 2025 – Present

- Led **AI feature integration** across core products — embedding LLM-backed assistants, automated content generation, and intelligent search into existing workflows via **API-based model integrations** with fallback, retry, and cost-control layers.
- Designed **prompt and context strategies** with structured outputs and schema validation, cutting malformed responses and eliminating downstream parsing failures in production.
- Accelerated delivery by adopting **agentic coding tools** (**Claude Code**, **Cursor**) into team workflow — scaffolding services, migrating legacy modules, and expanding test coverage, while enforcing human review gates on every AI-generated change.
- Optimized backend/API performance through caching, query tuning, and async workflows, materially improving p95 response times under load.
- Strengthened deployment reliability by introducing **CI/CD automation**, reducing manual release work and deployment failures.
- Improved uptime and mean-time-to-detect through structured error handling, monitoring, and **observability** instrumentation.
- Hardened security posture — remediated vulnerabilities, rebuilt authentication flows, and implemented role-based access control.
- Mentored junior engineers, owned onboarding documentation, and raised the bar on code review and testing standards (unit, integration, E2E).
- Partnered directly with clients and stakeholders to translate ambiguous business goals into scoped, shippable technical roadmaps.

Senior Software Engineer · CodingCops Feb 2022 – Jan 2025

- Owned **architecture, development, and deployment** of B2B platforms for the hospitality industry, serving multi-tenant production traffic.

- Migrated a monolith to **microservice architecture**, delivering a **35% performance improvement** and independent service scalability.
- Built AI-assisted product features on **GPT** and **Gemini** APIs — conversational interfaces, automated content workflows, and intelligent search — integrated through clean service boundaries rather than bolted onto the frontend.
- Automated the **CI/CD pipeline** for the flagship product with JUnit-based test coverage, reducing feature release time to near-zero manual effort.
- Authored **TDD** guidelines, technical documentation, user manuals, and GAP analyses adopted across the engineering org.
- Introduced git submodules to drive code reuse across the product line and cut duplicated maintenance.
- Drove code reviews, automated testing, and QA processes as a full participant in agile delivery.

Software Engineer · *TectSoft* Oct 2020 – Jan 2022

- Redesigned portal matching workflows to minimize turnaround time — a retail client saw a **20% increase in sales**.
- Troubleshot and stabilized **GitLab pipelines**, ensuring reliable delivery of release builds.
- Supported L1 teams through triage meetings and hands-on technical escalation.
- Delivered mobile applications for **Namibian government clients** — end-to-end ownership of design, architecture, and store publication.
- Led R&D on **Optical Character Recognition** with React Native and Tesseract, improving extraction accuracy on identity documents.
- Built a citizen police-verification platform from scratch, handling document verification, scanning, attestation, and gesture-based security.
- Configured on-premises TFS in line with local legal regulations for data-residency compliance.

SELECTED PROJECTS

AnyWork Agent — *React.js, Node.js, LLM APIs, Voice AI, Email Automation* · SMS Holdings

- Conversational **AI agent** that turns a natural-language prompt into a qualified lead list, then autonomously builds and launches an outreach campaign around it.
- Built **agentic tool-calling** workflows spanning lead discovery, campaign generation, **AI voice calling**, and automated email sequencing — with human approval gates before any outbound action.

MindMiner — *React.js, Node.js, AWS ECR/ECS/Fargate, Ethereum, LLM & Video Gen APIs* · CodingCops

- **AI-assisted invention platform** — users describe a problem and the system generates candidate solutions, then drafts structured **patent documentation** from them via LLM-backed pipelines.
- Integrated **AI video generation** for automated multimedia explainers of each invention, alongside prompt-driven content generation across the product.
- Web3 layer with wallet balances, on-chain burns, **ERC-20 and NFT transfers**, crowdfunding contracts, and token minting — secure multi-step authenticated transactions with blockchain event verification.

Smart System — *React Native, React.js, Node.js, GitLab, AWS* · SMS Holdings

- Mobile and web platform for on-demand household services, with an integrated admin panel for customer and partner monitoring and in-app commissioning support.

IVY — *Ruby on Rails, React.js, Hanami, Heroku, Datadog, Redis* · CodingCops

- Hospitality platform covering guest booking, task orchestration, SMS notifications, and an activity management system that improved staff task completion.

Smeisty — *React.js, Node.js, Ethereum, MetaMask, OpenSea* · TectSoft

- NFT management platform handling contract addresses and wallet transactions, optimized for **speed and cross-device responsiveness**.

EDUCATION

Bachelor of Science in Software Engineering — Lahore Garrison University, Lahore, Pakistan · CGPA: 3.4 / 4.0